MILITARY SPECIFICATION
PUMP, EJECTOR, AIR VEHICLE FUEL,
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense,

1.0 SCOPE

This specification defines minimum requirements for air vehicle fuel ejector pumps, regardless of application (transfer, boost, scavenge, etc.), configuration (multiple nozzle, multiple stage, variable geometry, etc.), or auxiliary features (check valves, motive flow shut off valves, negative "G" provisions, etc.).

2.0 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-S-735	Standard Test Fluids, Hydrocarbon
VV-G-109	Gasoline, Unleaded
PPP-B-591	Boxes, Fiberboard, Wood Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-S-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2915

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Military (cont'd)

MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-C-6021	Castings, Classification and Inspection of
MIL-C-7024	Calibrating, Fluids, For Aircraft Fuel System Components
MIL-P-7105	Pipe Threads, Taper Aeronautical National Form, Symbol Anpt., General Requirements For
MIL-S-7742	Screw Threads, Standard Optimum Selected Series; General Specification For
MIL-F-8615	Fuel System Components; General Specification For
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter; General Specification for
MIL-P-15024	Plate, Tags and Bands for Identification of Equipment
MIL-A-21180	Aluminum-Alloy Castings, High Strength
MIL-N-25027	Nuts, Self-Locking 250°F, 450°F, 800°F, 125 KSI FTU, 60 KSI FTU, and 30 KSI FTU
MIL-R-25988	Rubber, Fluorosilicone Elastomer, Oil and Fuel Resistant
MIL-F-27351	Fluid, Calibrating, High Flash Point Aircraft Fuel Systems Components
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-R-83248	Rubber, Fluorocarbon Elastomer, High Temperature

STANDARDS

Federal

FED-STD-151	Metal, Test Methods

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specification and Standards Order of Precedence for Selection of
MIL-STD-889	Dissimilar Metals
MIL-STD-1523	Age Controls of Age-Sensitive Elastomeric Material.
MS20995	Wire, Safety or Lock
MS29512	Packing Preformed Hydrocarbon Fuel Resistant O-ring
MS29513	Packing Pre-Formed Hydrocarbon Fuel Resistant O-ring
MS33588	Nuts and Plate Nuts, Self-Locking, ACFT Design & Usage Limitations of
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal.

American Society for Testing and Material

ASTM D 1655	Aviation Turbine Fuels

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3.0 REQUIREMENTS

3.1 First Article - This specification makes provisions for First Article testing (see 4.4).

3.2 Model Specification - An ejector model specification conforming to the appendix shall be submitted by the manufacturer for approval.

3.2.1 Existing Ejector Models - Prior to the preparation of a new ejector model specification, the airframe manufacturer shall review the characteristics of existing standard ejector models for utilization, in order to eliminate duplication of models.

3.2.2 Drawings. Drawings shall be in accordance with MIL-D-1OOO at the time and place of first article approval. The data shall include but not be limited to: top drawing, assembly drawing and the data delineated thereon, e.g., specifications, manufacturing instructions, etc.

3.2.3 Wueight. The allowable dry weight shall be stated in the item specification.

3.3 Definitions - The following terms are defined and used in this specification:

Motive Flow	- High pressure fuel supplied to nozzle of ejector pump
Wn	- Flowrate into motive flow port (kg/see)
Ws	- Flowrate into suction inlet (kg/see)
Wt	- Flowrate from discharge port (kg/see)
Pi	- Static nozzle pressure (kPa)
Pc	- Static inlet pressure (kPa)
Pd	- Static discharge pressure (kPa)
$\overline{Pi}$, $\overline{Po}$, $\overline{Pd}$	- Total pressure

3.4 Selection of Materials, Processes and Parts - Unless otherwise specified, all materials, processes and parts shall be selected in the order of precedence described in MIL-STD-143.

3.4.1 Commercial Parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable military standard parts. In any case, commercial utility parts, such as screws, bolts, nuts and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.
b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the manufacturer's drawings.

3.5 Materials - Materials and processes shall conform to applicable Government specifications . Materials or processes conforming to contractor's specifications may be used with approval of the procuring activity.

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3.5.1 Fungus-Proof - Materials that are nutrients for fungi shall not be used.

3.5.2 Metals - Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray or atmospheric conditions likely to be met in storage or normal service. The use of any protective coating that will crack or scale with age or extremes of climatic and environmental conditions shall be avoided. Metals shall be inspected in accordance with FED-STD-151. Magnesium, copper, and alloys thereof shall not be used in contact with fuel.

3.5.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5.2.2 Cadmium plating. Cadmium plating shall conform to QQ-P-416, Class 2, Type II. Cadmium plating shall not be used on surfaces which would normally be in contact with fuel(s) unless specifically approved by the procuring activity.

3.5.3 Castings - Castings shall be in accordance with the applicable class designated in MIL-A-21180 and MIL-C-6021. Elongation in any specimen cut from the casting or integrally cast with the part shall be unacceptable if less than 3%.

3.5.4 Synthetic Rubber

3.5.4.1 Age Controls - Age controls for synthetic-rubber parts shall be in accordance with MIL-STD-1523. Wherever possible, Elastomers not subject to age control shall be used.

3.5.4.2 Serviceability - All synthetic-rubber parts shall be readily replaceable with a minimum removal of attaching parts.

3.5*4.3 Packings - All 'O' ring packings shall conform to the dimensional requirements of MS 29513 or MS 29512 or MIL-R-25988/l or MIL-R-83248/l and elastomer compounds as approved by the procuring agency.

3.5.5 Fluids. The item specification shall state the principle fluid and other fluids expected to be encountered in use. Fuel system components shall normally be compatible with MIL-G-5572, MIL-T-5624, MIL-C-7024, MIL-F-27351, TT-S-735, W-G-109, ASTM-D-1655, Types A, A-1, and B, and other comparable fuels. In normal operation, these fuels may contain various quantities of dissolved and free water, salts, gases, some solid contaminants, such as sand and metal chips.

3.6 Design - The ejector shall be operable in any orientation (unless otherwise required by the model specification), providing the suction inlet is submerged in fuel. The general design and construction shall be in accordance with para 3.6.1, 3.6.2 and 3.6.3 of MIL-F-8615.

3.6.1 Inlet Screen - When an inlet screen is required by the model specification the ejector shall incorporate a screen with openings equivalent to No. 8 mesh, .58 to .74 mm (0.023 to 0.029 inch) diameter wire that shall completely

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cover the inlet port and shall be securely fastened on all sides to the ejector housing to prevent passage of foreign material into the ejector. The inlet screen shall be large enough to permit required maximum flow with acceptable pressure drop when 50 percent of the inlet screen is blocked.

3.6.2 Fitting Ends - Fitting ends, other than discharge and primary flow inlet ports, shall conform to MS 33649 and MS 33656, except as otherwise required by the model specification.

3.6.3 Mounting - Ejector mounting shall be tank or line mounted as specified in the model specification.

3.7 Construction - The ejector shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service use. No parts shall work loose in service.

3.8 Performance - The ejector shall satisfy the following performance requirements.

3.8.1 Calibration - The range of pressure ratio and flow ratio shall be as defined in a curve similar to Figure 1. Ref. Figures 2 and 3.

3.8.2 Pressure Loss - Pressure loss with the ejector functioning as a suction pickup with no motive flow supplied shall not exceed that specified by the model specification. The pressure loss referred to is static pressure loss.

3.8.3 Pump Down - Minimum pump down level shall be as specified by the model specification. Pump down level is defined as that fuel level at which Ws (refer to Figure 4) drops to zero.

3.8.4 Altitude - The ejector shall function satisfactorily when subjected to the altitude test (4.6.5) and shall be in accordance with the altitude performance test curves in the model specification.

3.8.5 Fuel Resistance - The ejector shall function satisfactorily when subjected to the fuel resistance extreme temperature test of Section 4.6.6.

3.8.6 Gravity Force. The performance during periods of inverted flight (negative G) or changes in gravity force shall be in accordance with the detail specification.

3.8.7 Endurance. In general, components whose life is time dependent shall have a minimum endurance requirement in accordance with the detail specification.

3.8.8 Water and ice. Water (or ice) shall not be a cause of a fuel system component failure. The water may be present in the fuel in the free state or dissolved. The component whose failure, due to water (and/or ice) could prevent transfer of fuel shall demonstrate its capability to satisfy aircraft system requirements in the presence of water in the fuel, either above freezing or below freezing.

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3.8.9 Environmental - There shall be no evidence of undue wear, corrosion, or any other damage that might adversely affect function, following exposure to any applicable environment specified herein or in the model specification.

3.9 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.10 Maintainability. Maintainability aspects of ejector pumps shall be in accordance with MIL-F-8615, paragraph 3.10.

3.11 Threaded connections.

3.11.1 Pipe Threads - Pipe threads shall conform to MIL-P-7105 and be used only on permanent closures.

3.11.2 Screw Threads - Screw threads shall conform to MIL-S-7742 and MIL-S-8879.

3.11.3 Locking of Parts - All threaded parts shall be locked by safety wiring, self-locking nuts in conformance with MIL-N-25027, cotter pins or other methods approved by the procuring activity. Safety wire shall have a minimum diameter of .81 mm (0.032 inches) and shall conform to MS 20995. Self-locking nuts shall be used in accordance with MS 33588. Where loosening or disengagement of the self-locking nut could result in the nut or other parts entering the fuel system, approval of the installation shall be obtained from the procuring activity. Staking and the use of lockwashers shall not be permitted.

3.12 Dimensions - Complete outline dimensions for the ejector shall be as specified in the model specification.

3.13 Mass - The mass of the ejector shall be as specified in the model specification.

3.14 Finish

3.14.1 Protective Treatment - All aluminum-alloy parts shall be anodized in accordance with MIL-A-8625, or adequately treated in some other acceptable manner for corrosion prevention. Stainless steel parts shall be passivated per MIL-S-5002.

3.14.2 Paint - Paint shall not be used except for color code identification per MIL-F-8615.

3.15 Identification of Product - Equipment, assemblies, and parts shall be permanently marked for identification in accordance with MIL-STD-130. The marking shall include the following minimum information:

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Pump, Ejector
Specification MIL-_____
Manufacturer's Code Identification
Manufacturer's Part No.
Serial No.

If a separate nameplate is used, it shall be a stamped or etched plate (aluminum or corrosion resistant steel) conforming to MIL-P-15024.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Inspection Responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other Inspection facilities and services acceptable to the Government. Inspection and quality control records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in this specification or the model specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests - The inspection and testing of the ejector shall be classified as follows:

a. First Article Tests (4.4)
b. Acceptance Tests (4.5)

4.3 Test Conditions

4.3.1 Atmospheric Conditions - Unless otherwise specified, all test of this specification shall be conducted at a room and fuel temperature between 16°C and 32°C (60°F and 90°F), and a barometric pressure of 95 to 108 kPa (28 to 32 inches of mercury) and a relative humidity of 80 percent or less. Where conditions are substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.3.2 Quantity of Fuel and Submergence - Unless otherwise specified, the quantity of fuel shall be sufficient to completely cover the ejector inlet and to prevent malfunctioning incidental to an excessive rate of recirculation.

4.3,3 Ejector position - Unless otherwise specified, the pump mounting shall simulate installation in the air vehicle and orientation shall simulate the normal vehicle operating attitude. The details of installation and orientation shall be included in the model specification.

4.3.4 Test Setup - The test set up for all performance tests shall be similar to that shown in Figure 5, and reference thereto shall include the ejector and fuel in the system. The pump installation shall simulate the actual aircraft installation to the degree required by the model specification.

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4.4 First Article Tests

4.4.1 First Article Test Sample Tested by the Supplier - The supplier shall subject the ejector to the preproduction test specified in Table I and additional tests as required by the Model Specification. The number of test articles and the order of tests shall be in accordance with Table I.

TABLE I. First Article Test Program.

	INSPECTION	REQUIREMENT	TEST	TEST ARTICLE 1	TEST ARTICLE 2
1	Examination of Product	3.6	4.6.1	X	X
2	Calibration	3.8.1	4.6.2	X	X
3	Suction Pressure Loss	3.5.5 and 3.8.2	4.6.3	X	X
4	Pump Down	3.5.5 and 3.8.3	4.6.4	X	X
5	Altitude	3.5.5 and 3.8.4	4.6.5	X	X
6	Fuel Resistance Extreme Temperature, Icing	3.8.5	4.6.6	X	
7	Accelerated Corrosion	3.8.5	4.6.7	X	
8	Endurance *	3.8.7	4.6.8	X	
9	Contaminated Fuel	3.5.5	4.6.9	X	
10	Water and Icing	3.8.8	4.6.10	X	
11	Negative "G" Cycling*	3.8.6	4.6.11		X
12	Vibration	3.8.9	4.6.12		X
13	Mechanical Shock	3.8.9	4.6.13		X
14	Mechanical Load	3.8.9	4.6.13		X
15	Maintainability	3.10	4.6.14	X	X

* If applicable due to valving, etc. being an internal part of the ejector.

NOTES:

a. Inspections 2, 3, 4 and 5 shall be repeated at the end of the tests for each test article prior to disassembly and inspection.

b. Inspections 6, 7 and 8 shall be conducted in the order listed.

c. A third test article is optional for sharing the inspections of test article 2.

4.4.2 Test Report - After the supplier completes the preproduction tests, he shall prepare a preproduction test report and furnish copies of the report as required to the procuring activity with the following minimum information:

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a. Report all tests, graphically presented when possible, together with a detailed statement indicating compliance, or extent of compliance with all requirements of this and the detail specifications referring specifically to paragraph numbers. Wherever a requirement is considered to be not applicable, the report should so state.

b. Summary of all tests.

c. Diagrams of all test setups.

d. Outline and description of test and test conditions.

e. Copies of test log sheets.

f. Photographs as required.

g. Assembly and cross-sectional drawings.

4.4.3 Test Sample for the Procuring Activity - All items subjected to test shall be available for submittal to the procuring activity for inspection and test for a period of at least two years following submittal of the test report. Additional test items shall be submitted as required by the contract or order.

4.5 Acceptance Tests - Acceptance tests shall consist of the following individual tests to which each ejector shall be subjected (the fuel shall be that specified in the model specification).

a. Examination of Product (4.6.1)

b. Calibration (test per 4.6.2.1 shall be performed at a single point. The nozzle pressure and flow ratio shall be specified by the detail specification. The test shall be performed only if required by the detail specification).

4.6 Test Methods

4.6.1 Examination of Product - The units shall be clean and free of any contaminants, oil, grease, preservatives, or any material not specified on the applicable assembly drawings. Each unit shall be inspected and certified to be in accordance with the applicable drawings and specifications.

4.6.2 Calibration

4.6.2.1 Pressure/Flow - Operate the ejector at the following conditions: Reid vapor pressure and max inlet pressure as specified in the model specification. The pump shall be tested at those nozzle pressures required by the model specification.

a. At each nozzle pressure (Pi) condition, discharge pressure (Pal) shall be varied over the complete operating range of the ejector.

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b. Sufficient data shall be obtained and curves plotted to demonstrate compliance with the model specification.

4.6.3 Suction Pressure Loss - (Perform only if required by the model specification] The ejector shall be installed in a fuel tank to simulate aircraft installation, with the nozzle port (Pi) blocked. The flow rate through the ejector shall be varied from O to design value by varying discharge pressure (Pal) at a constant secondary flow inlet port pressure (Po). The pressure change (Pd-Po is to be recorded and plotted. The pressure drop shall be as specified in the model specification.

4.6.4 Pump Down - The ejector discharge port shall be closed. Po and Pi shall be adjusted as specified in the model specification. The discharge port shall be opened, starting the ejector. The liquid level shall be permitted to drop as the ejector operates. The point at which Wt equals Wn is the minimum pump down level. This level is to be specified in the model specification.

4.6.5 Altitude - The ejector pump shall function satisfactorily when subjected to both transient altitude tests and steady state altitude test. The Transient Test varies the pressure altitude in the test tank from sea level to maximum obtainable within a time schedule, simulating air vehicle climb to altitude with maximum obtainable within a time schedule, simulating air vehicle climb to altitude with maximum Reid Vapor Pressure (RVP.) fuel (least fuel weathering). The Steady State Test holds the pressure altitude constant while checking the ejector pump performance, simulating the air vehicle at constant altitude with maximum RVP fuel.

a. Transient Test - This test shall be conducted with a sample of the applicable fuel with maximum RVP (as defined by the applicable fuel specification)* in the quantity required by the model specification. Fuel temperature shall be maintained at 57°C (135°F) unless otherwise specified. Total nozzle pressure (Pi) shall be held at the specified optimum design point. Total discharge pressure (Pal) shall be decreased at each altitude until limit flow occurs. Sufficient test data shall be taken to define the limit flow ratio at each altitude called for in the model specification.

b. Steady State Test - This test shall be conducted with a fresh sample of the applicable fuel with maximum RVP (as defined by the applicable fuel specification)* in the quantity required by the model specification. Test parameters PO, (Pi - Po} and fuel temperature shall be the same as test (a) of this section. A climb to the maximum obtainable pressure altitude for the RVP and test fluid temperature shall be made from sea level within the time required in the model specification. At the test altitude Pd shall be varied from its maximum to minimum value. This sequence shall not exceed 120 seconds. Identical sequences shall be taken at altitudes from maximum obtainable down to sea level in 3,000 meter [9,840-foot) increments starting at the service ceiling and progressing down to sea level.

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* NOTE

If difficulty is encountered in obtaining maximum Reid Vapor Pressure fuel as defined by fuel specification, it is permissible to mix a more volatile fuel, such as JP-3 or AvGas, with the test fuel until the correct RVP is obtained.

4.6.6 Fuel Resistance, Extreme Temperature and Icing Tests - The fuel resistance, extreme temperature, and icing tests shall be conducted in accordance with the appropriate tests specified in MIL-F-8615.

4.6.7 Accelerated Corrosion - (This test may be abbreviated or deleted when it can be demonstrated that all materials Individually and in contact in the assembly, are not subject to any manner of corrosive deterioration, whether chemical, electrolytic or stress accelerated.)

The complete pump with all fuel ports open, shall be submerged in a saturated salt solution. The unit shall then be drained for 30 seconds to remove trapped salt solution. The component shall be placed immediately in a test chamber maintained at a temperature of 30° ± 2°C (86° ± 4°F) with a relative humidity of 100 percent for a period of 1200 seconds. Upon completion of the humidity exposure period, the unit shall be placed in an air oven maintained at a temperature of 55° to 60° (131°F to 140°F) for a period of 1200 seconds. The component shall not be operated again until a total of 60 immersions, high humidity, and drying cycles have been completed. Immediately after completion, the component shall be washed with warm water, dried, wetted with fuel, and actuated as in normal service.

4.6.8 Endurance - Unless otherwise specified in the model specification, the endurance test shall be as specified in MIL-F-8615.

4.6.9 Contaminated fuel test. This test is applicable to all components required to perform the endurance test per MIL-F-8615, paragraph 4.5.12.

4.6.10 Water and icing. Tests shall be accomplished to demonstrate performance as required by the item specification. The tests shall be conducted as specified in paragraph 4.5.17 through 4.5.17.3 of MIL-F-8615, and in accordance with paragraph 3.8.8 Water and ice requirement of this specification.

4.6.11 Negative G Cycling Test (Applicable to pumps with negative "g" inlet valving only.) - With ejector mounted and operating at the optimum design point, the ejector shall be alternately subjected to 1 “g” normal and 1 “g” negative at a rate of one cycle per minute for 500 cycles. At any time during the cycling, performance shall be as required by the model specification. The tank shall be arranged so that the upper inlet is uncovered for the plus one “g” and the lower inlet is uncovered during the negative one “g” operation. The inlet valve shall operate smoothly from normal to negative “g” position in not more than 3 seconds after reversal of the "g" force.

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4.6.12 Vibration - Vibration tests shall be conducted in accordance with the model specification.

4.6.13 Strength. The ejector pump shall be subjected to mechanical shock per paragraph 4.5.15.1 and mechanical load test per paragraph 4.5.15.2 of MIL-F-8615.

4.6.14 Disassembly and inspection. There shall be no evidence of deterioration or corrosion that might affect performance of the ejector when disassembled and inspected at the conclusion of the tests, specified in section 4.

5.0 PREPARATION FOR DELIVERY

5.1 Preservation and Interior Packaging

5.1.1 Level A - Ejectors shall be Preserved and packaged one each in accordance with Method II of MIL-P-116.

5.1.2 Level B - Ejectors shall be preserved and packaged in accordance with Method 1A of MIL-P-116.

5.1.3 Level C - Ejectors shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing

5.2.1 Level A - Packing For Overseas Shipments - Units preserved and packaged as specified in 5.1.1 or 5.1.2 shall be packed in overseas type exterior containers conforming to PPP-B-591, PPP-B-621, PPP-B-636, PPP-B-601, or PPP-B-621. Exterior containers shall be of uniform size and shape and shall be of minimum cube and tare, consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable specification or appendix thereto.

5.2.2 Level B - Limited Packing (For Domestic Shipment) - Units preserved and packaged as specified in 5.1.1 or 5.1.2 shall be packed in domestic-type exterior containers conforming to PPP-B-591, PPP-B-601, or PPP-B-621. Exterior containers shall be of uniform size and shape and shall be of minimum cube and tare, consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable specification or appendix thereto.

5.2.3 Level C - Minimum Packing (For Domestic Shipment with Immediate Use at Initial Destination - Packages which require over-packing for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall conform to the Uniform Freight Classification Rules or regulations of the other common carriers as applicable to the mode of transportation.

5.3 Marking of Shipments - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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5.3.1 Reinspection Dates - Reinspection date marking shall be added as specified by the procuring activity.

6.0 NOTES

6.1 Intended Use - The ejectors covered by this specification are intended for use in air vehicles under condition specified in the model specification.

6.2 Ordering Date - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. The testing activity (see 4.4).

c. Applicable level of preservation, packaging, and packing (see 5.0).

d. Reinspection date marking (see 5.3.1).

6.3 Contract Data Requirements - Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.2.2	Drawings and test report	DI-E-6105 and DI-E-6108
4.4.2	Test report	DI-T-2072

6.4 Data - For the information of Contractors and Contracting Officers, any of the data specified in (a) applicable documents listed in Section 2 of this specification, or (b) referenced lower tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed in DD Form 1423 (Contractor Data Requirements List) which shall be attached to and made part of the contract or order.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Preparing Activity:
Navy - AS
(Project No. 2915-0095)

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EJECTOR TYPE FUEL PUMPS
AIRCRAFT FUEL

APPENDIX

10.0 SCOPE

10.1 This appendix establishes the form to be used by manufacturers in the preparation of an aircraft fuel ejector pump model specification.

20.0 APPLICABLE DOCUMENTS

20.1 The following document, of the issue in effect on date of invitation for proposals, forms a part of this appendix to the extent specified herein:

SPECIFICATIONS

Military

Pump, Ejector, Aircraft Fuel, General Specification for. (Copies of specifications, standards, drawings and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as' directed by the contracting officer.)

30.0 APPLICATION

30.1 A complete aircraft fuel ejector pump model specification, in accordance with the outline and instructions for preparation as specified herein, shall be prepared for each specific model. New ejector model designations will be assigned by the Government. When a new model designation is assigned, a new ejector model specification shall normally be prepared with a new specification number assigned thereto by the unit manufacturer. It is desired that no new ejector model designation be implemented by appendixes, or variants to an existing model specification, and, in no case, shall amendments be used for this purpose. No change to an ejector model specification shall be submitted to the procuring activity by means of amendments or revision pages prior to the time when the pump model specification is approved and becomes a part of a contract. Revisions by amendment form to an approved pump model specification, which has been released and is forming a part of a contract, will be acceptable to the procuring activity. Each amendment shall be approved, and shall include and supersede the previous amendment.

30.2 The headings and numbering of sections and paragraphs herein correspond to those of the basic specification for the specific data needed only in the model specification. When departures are necessary from the requirements of this appendix and of the basic specification, the details of such departures shall be stated as specific requirements bearing the same section and paragraph heading and numbering as in the basic specification.

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30.3 Parenthetical sentences, phrases, and words are included herein for the guidance of the pump manufacturer for insertion of proper information and data related thereto, in connection with the preparation of the model specification. Parenthetical statements shall not be copied verbatim in the model specification.

30.4 The specification number shall be the number assigned by the pump manufacturer. When revisions are made, they will be designated by the use of a dash and a letter following the number, with a revision date therefore, which shall be shown on page 1 only. Only the specification number, and the revision suffix letter, if applicable, shall be shown on the subsequent pages.

30.5 For purposes of permitting -preliminary evaluation of a proposed ejector design, or for release of approved ejector performance characteristics in connection with an aircraft or aircraft equipment design competition, the ejector manufacturer may submit a preliminary model specification to serve until superseded by a complete coordinated model specification which will be required for a production contract.

30.5.1 The preliminary model specification shall be prepared in accordance with the requirements stipulated herein.

40.0 MODEL SPECIFICATION

40.1 The form and description of the model specification follow:

(Number and title - The number and title shall be shown as follows.)

SPEC NO. _____
(DATE) _____ _____
(A) REVISED _____
(B) REVISED _____(DATE)

MODEL SPECIFICATION
PUMP, EJECTOR, FUEL BOOSTER, AIRCRAFT

(Insert service type and model designation if assigned)

(NAME OF CONTRACTOR)

(CONTRACTOR'S MODEL DESIGNATION)

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1.0 SCOPE

1.1 Scope - This specification covers the requirements for an aircraft fuel booster (or transfer) ejector pump. The pump shall meet all requirements herein and the requirements of MIL-P-85315, General Ejector Pump Specification.

1.2 Classification - The fuel booster (or transfer) ejector shall be of the following _____ (insert the contractor's model designation; also include a brief description of the ejector and its operation.)

2.0 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-85315 Pump, Ejector, Air Vehicle, Fuel, General Specification for

MIL-G-5572 Gasoline, Aviation; Grades 80/87, 91/96, 100/130, 115/145

MIL-T-5624 Jet Fuel, Grades JP-3, JP-4, and JP-5

MIL-T-83133 Jet Fuel, Grade JP-8

STANDARDS

Military

(list all additional Government publications not covered this specification which are applicable to this pump).

3.0 REQUIREMENTS

3.8.1 Calibration - The ejector shall have the capacities, rated flows, discharge pressures, specified by the Contractor, using fuel in accordance with MIL-P-85315. (Insert the applicable specification number)

3.11 Dimensions - A drawing shall be furnished with this model specification, showing the outline and mounting dimensions for the ejector.

3.12 Mass - The weight of the ejector, complete as a working assembly, shall not exceed _____ kg.

4.0 QUALITY ASSURANCE PROVISIONS

4.6.2 Calibration - The ejector shall be operated over its entire permissible range of flow rates. (Data covering discharge pressure shall be recorded.)

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4.6.3 Pressure Loss - With the motive flow zero (nozzle port blocked), fuel shall be drawn through the idle ejector by auxiliary means at flow from 0 to rated flow in kilograms per second.

4.6.4 Pump Down - The pump down level shall be no more than meters above the suction inlet.

4.6.5 Altitude - The ejector shall conform to the altitude climb curves as specified by the airframe manufacturer for the aircraft requirements and shall be within the minimum requirements of the altitude climb curves as shown in Figure 5. All altitude climb tests shall be conducted with fuels conforming either to MIL-G-5572 or MIL-T-5624, as applicable.

4.6.8 Vibration - (The Contractor shall specify the procedures used in vibration testing).

5.0 PREPARATION FOR DELIVERY

5.1 The preparation for delivery shall be in accordance with (this specification no.)

6.0 NOTES

6.1 Intended Use - (The Contractor shall give a brief description of the intended application of the ejector in the aircraft).

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*EFFICIENCY = $\overline{N}$ Ø

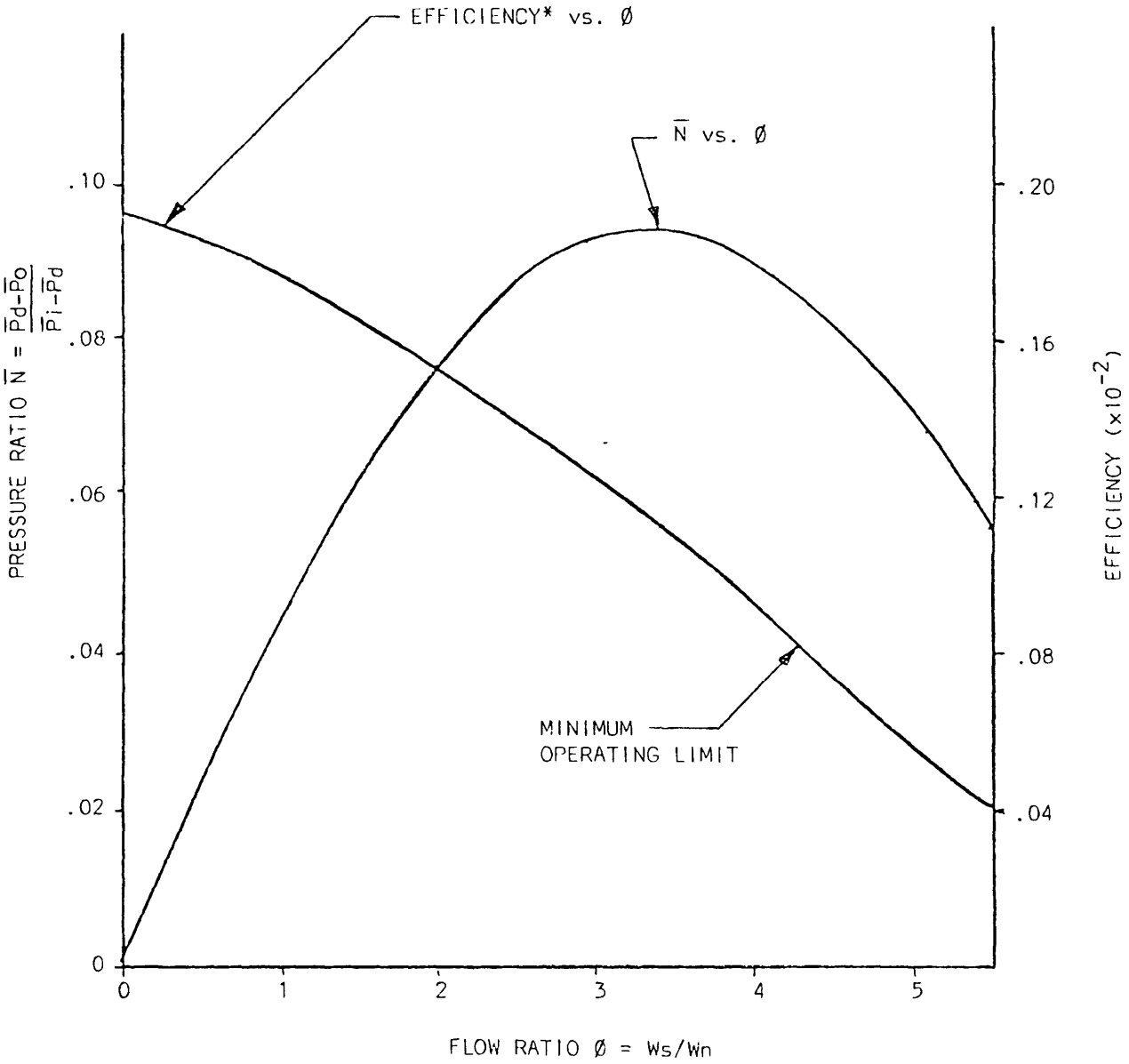


Figure 1. Typical Sea Level Performance Curve For Ejector Transfer Pump

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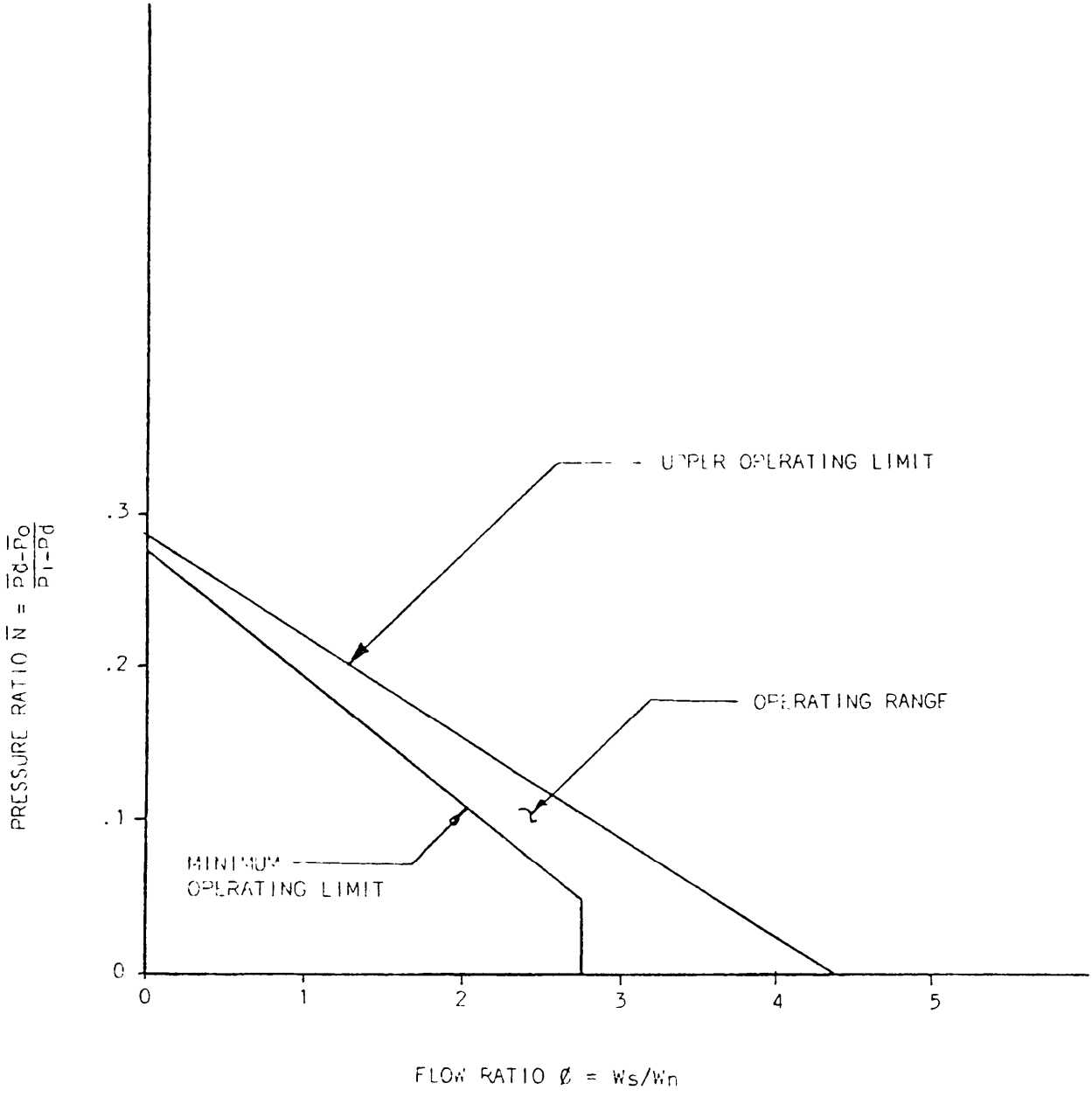


Figure 2. Typical Performance Curve for Ejector Boost Pump With Negative "G" Capability

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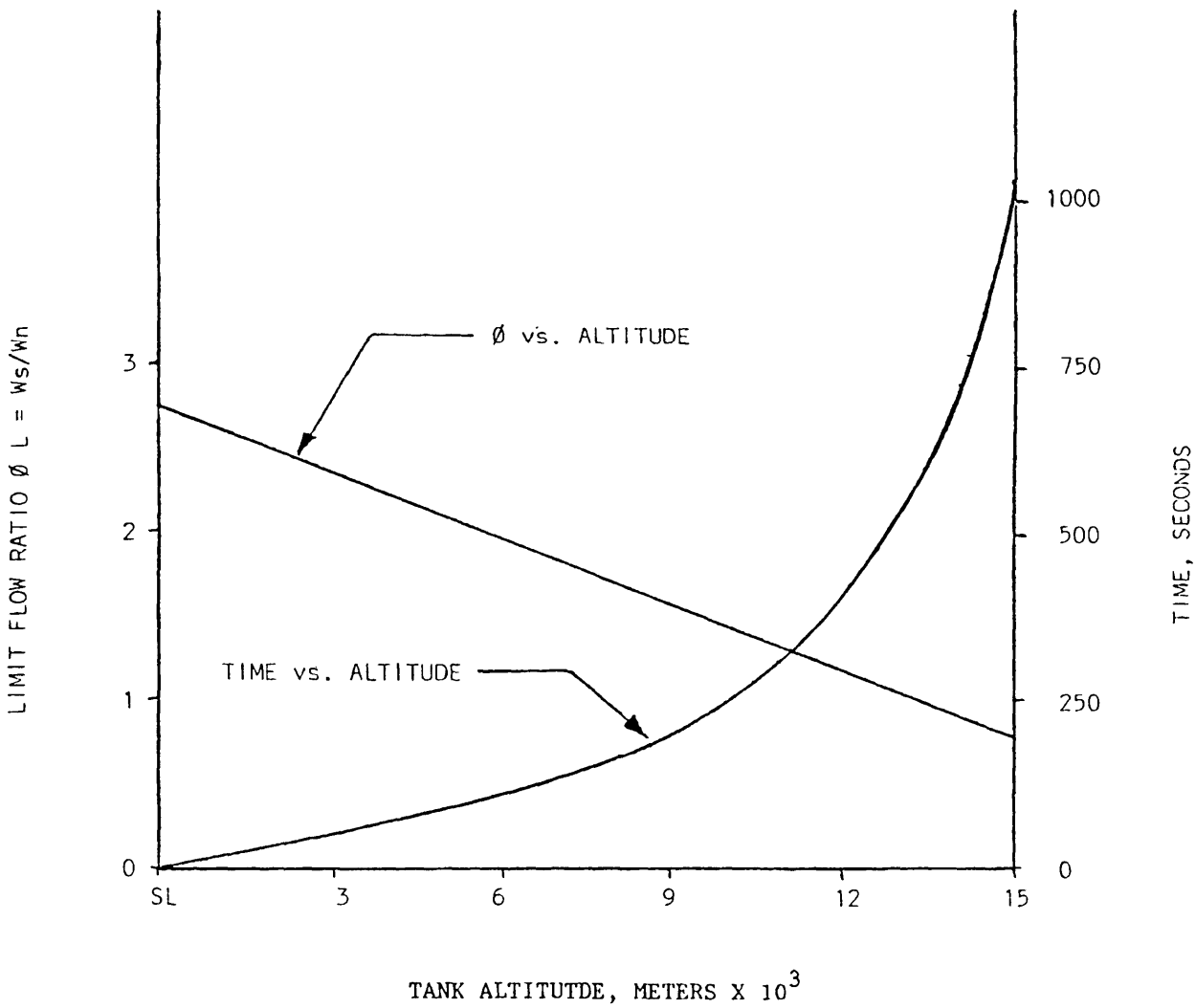
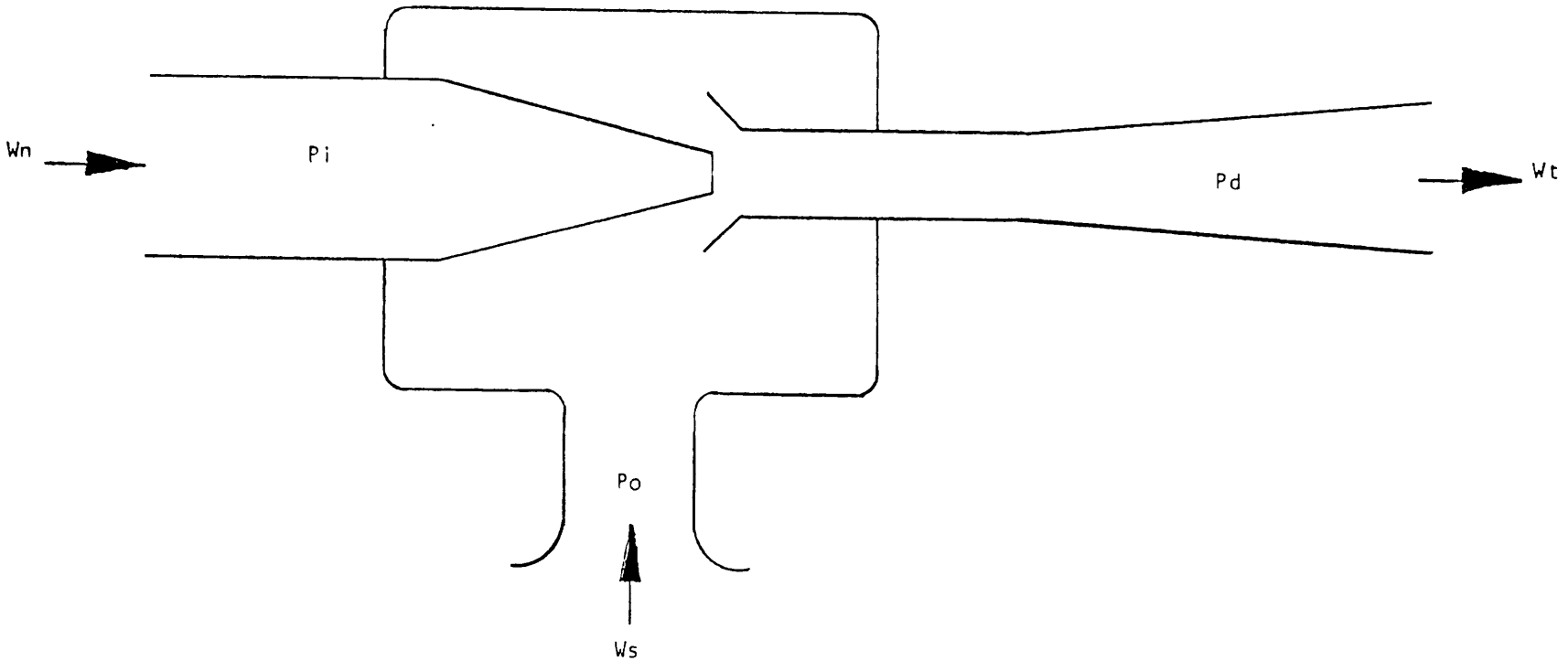


Figure 3. Typical Altitude Performance Curve

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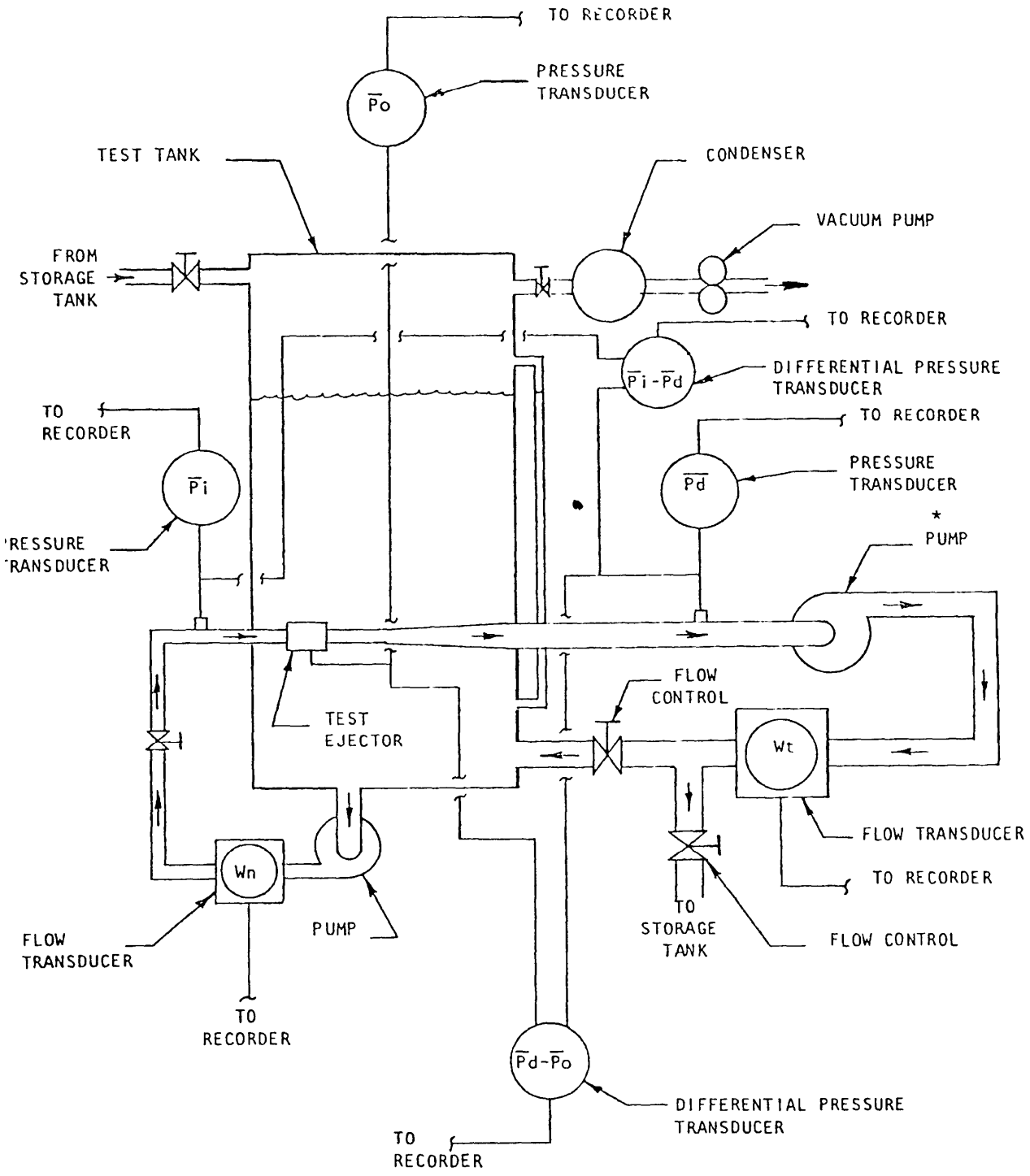


Wn	---	Flowrate into motive flow port (kg/sec)	$\emptyset$ = Ws/Wn = (Wt - Wn)/Wn termed "Flow ratio"
Ws	---	Flowrate into suction inlet (kg/sec)	
Wt	---	Flowrate from discharge port (kg/sec)	$\bar{N} = (\bar{P}d - \bar{P}o)/(\bar{P}i - \bar{P}d)$ termed "Pressure ratio"
Pi	---	Static nozzle pressure (kPa)	$\emptyset$ = Ws/Wn = (Wt - Wn)/Wn termed "F"
Po	---	Static inlet pressure (kPa)	
Pd	---	Static discharge pressur (kPa)	$\bar{N} = (\bar{P}d - \bar{P}o)/\bar{P}i - \bar{P}d)$ termed "Pressure ratio"
$\bar{P}i$	---	Total nozzle pressure *	
$\bar{P}o$	---	Total inlet pressure (kPa)*	
$\bar{P}d$	---	Total discharge pressure (kPa)*	

* Total Pressure - static + dynamic pressures. (Dynamic pressure in many pumps may be so small that it may be disregarded.

Figure 4. Ejector Parameters

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* If needed to achieve desired discharge flowrate.

Figure 5. Typical Test Setup Schematic

* If needed to achieve desired discharge flowrate.

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